



Measuring China's monetary policy uncertainty and its impact on the real economy

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ABSTRACT

This paper proposes a measure of China's monetary policy uncertainty (MPU) based on a Bayesian MCMC estimation procedure, investigates the economic impact of MPU, and outlines the impact channels. Empirical findings suggest that MPU depresses China's economy via higher credit risk and lower economic growth. Moreover, the decrease in output is amplified and made more persistent when the economy is simultaneously exposed to higher credit risk. Because China has experienced massive credit expansion over the past decades, the economy is currently more vulnerable to the increase in MPU than it ever was.

1. Introduction

Concerns about policy uncertainty have increased significantly in the aftermath of global financial crisis, especially in recent years when the average global uncertainty index hits a new high. Since the pioneering work of Bloom (2009), a rapid-expanding literature is devoted to measuring uncertainty and its impact on economic activities (see, e.g., Baker et al., 2012; Fernández-Villaverde et al., 2015; Jurado et al., 2015; Demir and Ersan, 2017). “Uncertainty is the defining characteristic of monetary policy landscape”,¹ and hence a growing number of studies are placing their attention squarely on monetary policy uncertainty (MPU) and its transmission mechanism (e.g., Mumtaz and Zanetti, 2013; Creal and Wu, 2017; Husted et al., 2019). So far, most researches are based on the evidence of advanced economies, and very little is known about MPU in developing economies. In this paper, we fill this gap by studying MPU in China, the world's largest developing economy.

We propose a measure of MPU in China by adopting a Bayesian MCMC approach to estimate China's MPU based on conditional volatility of the unforeseeable component of monetary supply. Though monetary policy in China is starting to resemble that of advanced economies (He and Wang, 2012; Fernald et al., 2014), significant differences remain in several aspects. First, China's monetary policy considers goals beyond the conventional output and inflation targets, particularly in recent years when the People's Bank of China (PBoC) have tried to enhance financial stability by limiting risky lending in the context of stable money supply.² Meanwhile, the number of instruments adopted by the PBoC keeps growing, making it hard for outsiders to measure the stance of

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¹ Quoted from Alan Greenspan, former Chairman of the Federal Reserve

² Prevention and control of financial risk has become a new priority for the PBoC since the fifth National Financial Work Conference of China in 2017.

monetary policy. Secondly, money supply has been one of the most crucial intermediate target for PBoC over the past two decades, which contrasts sharply against the emphasis on interest rates targets in advanced economies. Compared to interest rate, the growth rate of money supply provides a better characterization of the monetary policy stance in China, because it responds more promptly to inflation rate and real economy (Chen et al., 2018). Interest rate, as a monetary policy tool, is less prominent in China also because the interest rate channel of monetary transmission in China is in a nascent stage (Kamber and Mohanty, 2018). Therefore, we seek to capture China's MPU based on the time-varying volatility of money supply.

We also investigate the transmission mechanism of China's MPU by estimating its impact on credit risk and economic fluctuation in structural VAR models, while the transmission of monetary policy has been discussed in Chen et al. (2017), Yannick (2018) and Fang et al. (2019) etc. China has experienced rapid economic growth fueled by high levels of bank lending, causing global concerns about the risks and instability of China's financial system. Because policy risk is a fundamental source of financial risk (e.g., Song and Xiong, 2018), our paper contributes to the understanding of financial risk by documenting the effect of MPU on credit risk, a factor rarely considered in existing literatures. Our estimation suggests that: (1) The rise in MPU results in higher ratio of non-performing loans for China's commercial banks, thereby exacerbating credit risk. (2) Consistent with the existing literature on uncertainty (e.g., Baker et al., 2012; Bloom, 2014), the increase in MPU also leads to a decrease of output in China. (3) Moreover, the impact of MPU shocks on economic activities is amplified by credit risk.

The rest of the paper is organized as follows. Section 2 introduces our new measure for China's MPU. Section 3 discusses how MPU affects the economy based on bank-level regressions. Section 4 focuses on economic outcomes from the macro perspective. Section 5 concludes.

2. Measuring China's MPU

Following the definition of Jurado et al. (2015) on uncertainty, MPU can be measured as the conditional volatility of purely unforecastable component of the future value of money supply. This paper suggests the use of monetary policy rule (Chen et al., 2018) to describe how the PBoC manages the money supply. The money supply, measured by M2, is an important monetary policy tool in China. M2 consists of M1 and "near money", and the definition of "near money" can be slightly different among every country. In China, "near money" mainly comprises time and savings deposits in national currency of resident non-bank financial corporations and non-bank non-government sectors with the PBoC and banking institutions. The PBoC and central banks in advanced economies share the same goals of price stability and output growth, so Chen et al. (2018) include inflation rate and output variables in their monetary policy rule.

Apart from the conventional output and inflation targets, Chinese government, in recent years, are focusing more on stabilizing the economy. According to the speech of Xiaochuan Zhou in Washington DC, the former president of China's central bank claimed that currency value stability is also one of the key priorities of China's monetary policy. Shortly after, China endeavored to trade short-term deleveraging pains for the long-term health of the economy, and started a deleveraging campaign to contain systematic financial risk, which caused the PBoC to shift its tactic dramatically. Therefore, in our monetary policy rule, we also consider RMB exchange rate, which also reflects how global factors affect China, and macro leverage ratio as follows:

$$g_t = g_0 + \gamma_0 g_{t-1} + \gamma_1 (\pi_{t-1} - \pi^*) + \gamma_2 (y_{t-1} - y_t^*) + \gamma_3 rer_{t-1} + \gamma_4 lev_{t-1} + e_t, \quad (1)$$

where g_t is the growth rate of money supply, π_t is CPI inflation rate, y_t is GDP growth rate, e_t is the exogenous error terms following Chen et al. (2018). In addition, rer_t represents RMB exchange rate and lev_t represents macro leverage ratio. We use the real effective exchange rate index as the proxy for rer_t , and the detrended credit to GDP ratio³ as the proxy for lev_t , these data are released from Bank of International Settlements (BIS). At the end of each quarter t , the PBoC determines g_t for the next quarter in response to the output gap and the gap between the actual CPI inflation and the inflation target (π^*), as well as variations in exchange rate and macro leverage ratio. Following Chen et al. (2018), the annual inflation target π^* is set to 3.5%. The output gap is defined as the difference between the actual GDP growth rate and the growth target y_t^* set annually by the Chinese State Council. The parameters represent the responsiveness of monetary policy to the each target respectively.

Following Fernández-Villaverde et al. (2011), we augment the above quantity-based monetary policy rule with stochastic volatility:

$$e_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \quad (2)$$

$$\ln \sigma_t^2 = \rho_0 + \rho_1 \ln \sigma_{t-1}^2 + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2), \quad (3)$$

where the stochastic volatility measure, σ_t , is our proxy for monetary policy uncertainty. Our measurement of uncertainty is similar in spirit to Jurado et al. (2015) and Mumtaz and Theodoridis (2017). We estimate Eqs. (1)–(3) via Bayesians MCMC based on the quarterly data of M2 growth rate, real GDP growth rate and CPI inflation rate from 1996Q1 to 2018Q3. The quarterly data of M2 growth rate, real GDP growth rate and CPI inflation rate are obtained from the CEIC database, and we manually collected the GDP growth rate target from the website of Chinese State Council. The details of our MCMC algorithm are documented in the Appendix, where we also present diagnostic test results to show that our algorithm performs well.

As is shown in Table 1, the PBoC engages in countercyclical monetary policy, with money supply responding negatively to the rise

³ We run the HP Filter to remove seasonality (detrend) from the series of total bank credit to GDP ratio.

Table 1
Estimated parameters.

Parameter	$\hat{g}_0$	$\hat{\gamma}_0$	$\hat{\gamma}_1$	$\hat{\gamma}_2$	$\hat{\gamma}_3$	$\hat{\gamma}_4$	$\hat{\rho}_1$	$\hat{\rho}_0$	$\hat{\sigma}_\eta$
Mean	0.802	0.934	-0.223	-0.123	0.92	-0.086	0.215	0.236	1.257
std	2.355	0.042	0.11	0.071	1.889	0.025	0.004	0.004	0.019

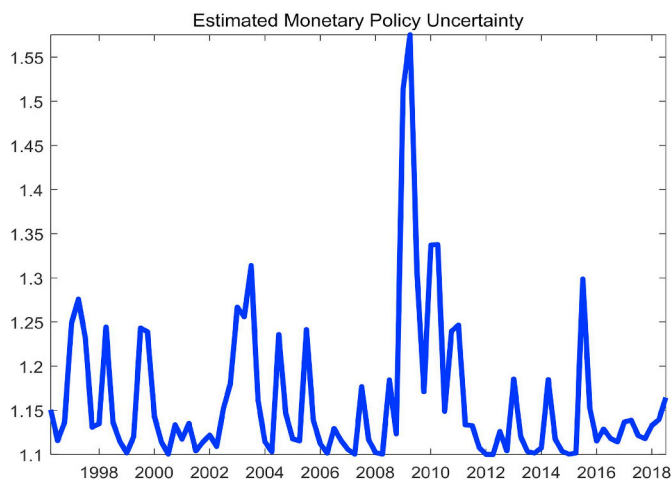


Fig. 1. China's monetary policy uncertainty.

in inflation and GDP growth. Moreover, for the purpose of maintaining financial stability, the PBoC increases money supply under the pressure of appreciation, tighten money supply as leverage grows, and vice versa. When money supply changes beyond the scope justified by conventional inflation and output targets and maintaining financial stabilization, we regard such excess movements as unpredictable components of monetary policy, whose conditional volatility can be the fundamental source of monetary policy uncertainty.

Fig. 1 indicates that China's MPU fluctuated wildly during the period of global financial crisis from 2008 to 2010. Before then, in 2007 and early 2008, controlling inflation was the key concern of China's monetary policy due to elevated levels of consumer price index (CPI) and the risk of housing bubbles. However, when the crisis broke out in 2008, the PBoC immediately switched its objective from containing inflation to stimulating output. The sudden and drastic change drove MPU to a record high. Loose monetary policy, together with the fiscal stimulus package in 2009, helped China gradually recover from the crisis. Later in 2010, the central bank switched back to prudent monetary policy, leading again to a spike in MPU. Since 2010, China's monetary policy has mostly remained prudent, and MPU has stayed at a low level, except for moderate rises during the “cash crunch” in 2013⁴ and the stock market crash in 2015.

3. Economic outcomes: Micro-level evidence

A well-established result in the literature is that uncertainty affects economic activities through the channel of financial frictions (see, e.g., Arellano et al., 2019; Christiano et al., 2014). In the context of the US economy, Gilchrist et al. (2014) show that an increase in uncertainty leads to significant widening of credit spread so as to compensate for default worries.

To answer the question of whether the credit channel matters for transimission of monetary policy uncertainty in China, we estimate the following model of credit risk:

$$\text{Creditrisk}_{it} = \alpha_0 + \alpha_1 \text{MPU}_{t-1} + \alpha_2 Z_{it-1} + \mu_i + \varepsilon_{it}, \quad (4)$$

where i is the index for individual banks. Credit risk, or the risk of default on debt, is measured by the non-performing loan ratio (NPLR) of China's commercial banks.⁵ Alternative measures for credit risk that we also explore, such as impaired loans ratio (IPLR), impaired including restructured and past due loans ratio (IRPLR), loan loss reserves ratio (LLRR) and net charge offs ratios (NCOR), are all positively correlated with the NPLR. The greater the NPLR ratio, the higher the credit risk. Our sample period is 2000–2014.

⁴ The MPU also rises in the aftermath of “cash crunch” in 2014, as interbank dismantling business are lassoed with stronger regulation.

⁵ NPLR is calculated by the summation of subordinate doubtful and loss loans divided by total loans of commercial banks. The data are manually collected from the CSMAR database, the Bankscope database and each bank's official website. We winsorize the data at 1% and 99%. The calculation of other proxies including IPLR, IRPLR, LLRR and NCOR follows the same rule.

Table 2
The effect of MPU on credit risks.

	NPLR	IPLR	IRPLR	LLRR	NCOR
MPU	9.566*** (1.456)	8.965*** (1.657)	24.247*** (5.219)	6.984*** (2.020)	5.961*** (1.418)
Size	0.089* (0.052)	−0.062 (0.068)	0.878 (0.703)	−0.052 (0.144)	0.019 (0.075)
ROE	−0.041** (0.018)	−0.012 (0.020)	0.024 (0.084)	−0.002 (0.020)	−0.009 (0.012)
LTD	0.000 (0.001)	0.000 (0.000)	−0.002 (0.002)	−0.001*** (0.000)	0.000 (0.000)
Age	−0.414*** (0.155)	−0.313** (0.157)	−0.981 (0.717)	−0.071 (0.284)	0.069 (0.174)
Loan	0.050 (0.560)	0.101 (1.107)	8.739 (5.530)	−1.952** (0.849)	2.435*** (0.926)
LG	−0.196 (0.158)	−0.092 (0.266)	−0.695 (0.598)	0.062 (0.299)	−0.464 (0.416)
M2	−0.210*** (0.038)	−0.250*** (0.048)	−0.504** (0.193)	−0.157** (0.060)	−0.154*** (0.039)
GDP	0.146 (0.100)	0.098 (0.065)	0.092 (0.186)	−0.132** (0.061)	−0.088** (0.044)
adj. R2	0.415	0.158	0.015	0.157	0.185
Obs.	505	501	505	535	451

Note: Coefficient estimates are provided in the top row and standard errors in parentheses. *, ** and *** indicate significance at the 10%, 5% and 1% levels (two-tailed test) respectively.

The vector of control variable Z_{it} includes bank-level variables: total assets (Size), return on equity (ROE), loan-to-deposit ratio (LTD), loan-to-asset ratio (Loan), growth rate of loans (LG) and age (Age) for bank i at time t . Bank-level individual effects are represented by μ_i . Also, Z_{it} controls for effects of macroeconomic variables by including seasonal adjusted GDP growth rate (GDP), money supply growth rate (M2), as well as unobservable time effects η_t . The calculation of explanatory variable MPU follows Section 2. We summarize the bank-level regression results in Table 2.

Table 2 shows that the growth of money supply reduces credit risk with statistical significance, so the convention of adopting loose monetary policy when default rate is high seems justifiable. Typically, China's monetary authority has the history of directly injecting liquidity to the bank system to help them solve bad loan problems. Over the past decades, the NPL ratio has been relatively high in China. The NPL ratio surged from 12.81% in 2002 to 34.18% in 2003 from where it plunged to 15.10% in the very next year, when 45 billion dollars were injected to the Bank of China and China Construction Bank⁶ by Central Huijin Investment Ltd.⁷

However, the uncertainty triggered by the switch of monetary policy stance significantly increases the level of banking credit risk, regardless of the choice of proxies for credit risk. Therefore, a surprise expansion of monetary policy with the aim of stimulating the economy is likely to cause side effects. The reason is that an unexpected monetary expansion itself adds policy-related uncertainty to the financial market, which corroborates the findings of Song and Xiong (2018).

Because it is practically impossible to maintain an ultra easy monetary policy regime for a long period of time, the eventual exit from such easing and associated rise in policy uncertainty inevitably inflict another round of economic damage. Therefore, loose monetary policy is only a partial solution for containing credit risk. To mitigate the effects of monetary policy uncertainty, the monetary policy authority should consider avoiding overactive intervention in the first place.

Result 1. : A rise in MPU increases the credit risk of China's commercial banks.

4. Economic outcomes: Macro-level evidence

4.1. A Bayesian structural VAR model

Vector Auto Regression (VAR) models have become one of the most popular tools for investigating the effects of monetary policy since the 1990s, but traditional maximum likelihood VARs are often over-parameterized and imprecisely estimated if data quality is poor. Because Bayesian methods offer an appealing solution to these issues, this section considers a Bayesian Structural VAR model of four macroeconomic variables of monetary policy uncertainty (MPU), GDP growth rate (GDP), inflation rate (CPI), and money supply (M2). Given that B_0 is invertible, the reduced form of SVAR can be written as:

$$Y_t = cB_0^{-1} + B_0^{-1}B_1Y_{t-1} + B_0^{-1}B_2Y_{t-2} + \dots + B_0^{-1}B_pY_{t-p} + u_t, \quad (5)$$

where $Y_t = \{MPU, GDP, CPI, M2\}$ is the vector of observed values of the four variables. The intercept is denoted as cB_0^{-1} , and $B_1B_0^{-1}$,

⁶ Those are two of the four biggest state-owned commercial banks in China.

⁷ A Chinese investment company owned by the government of the People's Republic of China

$B_2B_0^{-1} \dots B_pB_0^{-1}$ are the coefficient matrices. The error term $u_t = B_0^{-1}\omega_t$, where ω_t is independent across t with zero mean, can be regarded as the weighted averages of the mutually uncorrelated structural innovations (ω_t). We apply the Cholesky decomposition to the reduced form residual covariance matrix to obtain B_0^{-1} .

We place MPU as the first variable because MPU represents the unexpected change of monetary policy and is therefore independent of the other variables. Technically speaking, the calculation in Section 2 makes measured MPU orthogonal to GDP growth and inflation. The Bayesian Structural VAR model is estimated with two quarterly lags and with the Minnesota priors.⁸

The structural impulse responses are presented in Fig. 2, where responses to MPU shocks are shown in the first column. In line with existing literatures on uncertainty, China's GDP growth responds negatively to the rise in MPU, especially in the first period when the negative response reaches -0.15% . Afterwards, the negative effect lasts for another three quarters while decreasing gradually. As MPU depresses economic activities, the CPI also responds negatively to a positive MPU shock, and thus the PBoC tends to stimulate the economy by increasing the money supply, thereby increasing M2 with a lag. To be specific, the positive response of money supply peaks in the second period, one quarter after the negative response of output reaches highest level. Based on this evidence, when easing monetary policy to counter economic downturns, the PBoC should take into consideration the side effects of easing because unexpected easing leads to an increase in MPU.

Result 2. : A rise in MPU depresses economic activities.

In the decomposition of the GDP growth rate, we focus on the specific period that follows the 2008 global financial crisis when MPU rose dramatically. As indicated by Fig. 3, the contribution of a positive MPU shock to GDP growth rate was -0.95% in 2009Q2, while that of a positive M2 shock was 0.32% . Therefore, the effectiveness of loose monetary policy in stimulating the economy was significantly diminished by the MPU arising from a sudden growth of money supply.

4.2. A threshold VAR model

To explain how heightened uncertainty curtails economic growth, macroeconomists have emphasized the importance of wait-and-see flexibility in making firm's future investment while facing increased uncertainty. Apart from MPU, credit risk is also regarded as a fundamental source of uncertainty. Intuitively, firms may have to postpone investments because banks tend to hold back lending in response to the rise in both credit risk and MPU. The reduction in credit supply due to credit risk implies a wider credit spread.

Inspired by Bonciani and Van Roye (2016) that demonstrate how frictions in credit supply can amplify the uncertainty shocks, we seek to estimate whether the exposure to credit risk affects China's ability to withstand the impact of positive MPU shocks. Threshold VAR (TVAR) model offers a simple way to capture possible nonlinearities such as asymmetric reactions to shocks,⁹ thus we introduce the following TVAR model:

$$Y_t = \begin{cases} c_1 + B_1(L)Y_t + \varepsilon_{1t} & \text{if } s_{t-d} \leq s^* \\ c_2 + B_2(L)Y_t + \varepsilon_{2t} & \text{if } s_{t-d} > s^* \end{cases} \quad (6)$$

s_{t-d} denotes the threshold variable with lag d , and s^* is the threshold value. We choose credit risk to be the threshold variable, while s^* is the threshold value above which the economy belongs into the high credit risk regime. $B_1(L)$ and $B_2(L)$ are regime-dependent lag polynomial matrices, while c_1 and c_2 are the intercept terms. The regime-specific innovation ε_{1t} and ε_{2t} are normally distributed with zero mean.

Our choice of threshold variable s_t is the credit-to-GDP gap (credit gap), defined as the difference between the credit-to-GDP ratio and its long-term trend. Credit gap was first documented by Borio and Lowe (2002) as an early warning indicator (EWI) for non-performing loans and banking crisis. In a healthy economy, credit gap is usually between 2–10%. However, according to the BIS, China reported a credit gap of 23.5% in 2016Q1 which was apparently excessive. To study the impact of the credit gap, we add the credit risk variable to the SVAR system by redefining $Y_t = \{MPU_t, GDP_t, CPI_t, M2_t, s_t\}$. To address the threshold effect, we estimate the model using Gibbs sampling and Metropolis Hasting Algorithm but leave the estimation details in the Appendix.¹⁰

In Fig. 4, we plot the estimated probabilities of high and low credit risk regimes. In accordance with BIS practice, the estimated threshold value for credit gap is 11% in our TVAR model. Overall, credit risk in China appeared to be low before 2010, with the credit gap fell below 11% most of the time. However, after the rapid and persistent expansion of credit after the global financial crisis, the Chinese economy entered a high credit risk regime in 2013. By the end of 2017, the Central Economic Working Conference of China emphasized the importance of containing systematic financial risk, inducing a switch back to a low credit risk regime in 2018.

As Fig. 5 retains the result that GDP growth responds negatively to a positive MPU shock, it also offers crucial new insight in that the negative response of output is stronger and more persistent in a high credit risk regime. Meanwhile, under a low credit risk regime, the impact of MPU shock fades away immediately after the first quarter. When the economy is exposed to high credit risk, the

⁸ We use AIC and HC to select the lag order for the SVAR model, and we also compare different priors set up within this SVAR model, such as Normal Wishart, Independent Normal Wishart, Normal Diffuse, Dummy Observations etc. The reason we choose Minnesota is that it minimizes DIC (Deviance Information Criterion) value for the model

⁹ Also used in Balke (2000) in the analysis of state-dependent effects of monetary policy

¹⁰ Before estimating the TVAR model, we choose the optimal lag order of 2 based on the AIC. We then run the nonlinearity test for the TVAR model against linear VAR model by applying the LR test statistics proposed by Hansen (1999). Based on the LR test statistics, we choose the first-order lag of the credit gap as threshold variable and pick the 2-regime TVAR model.

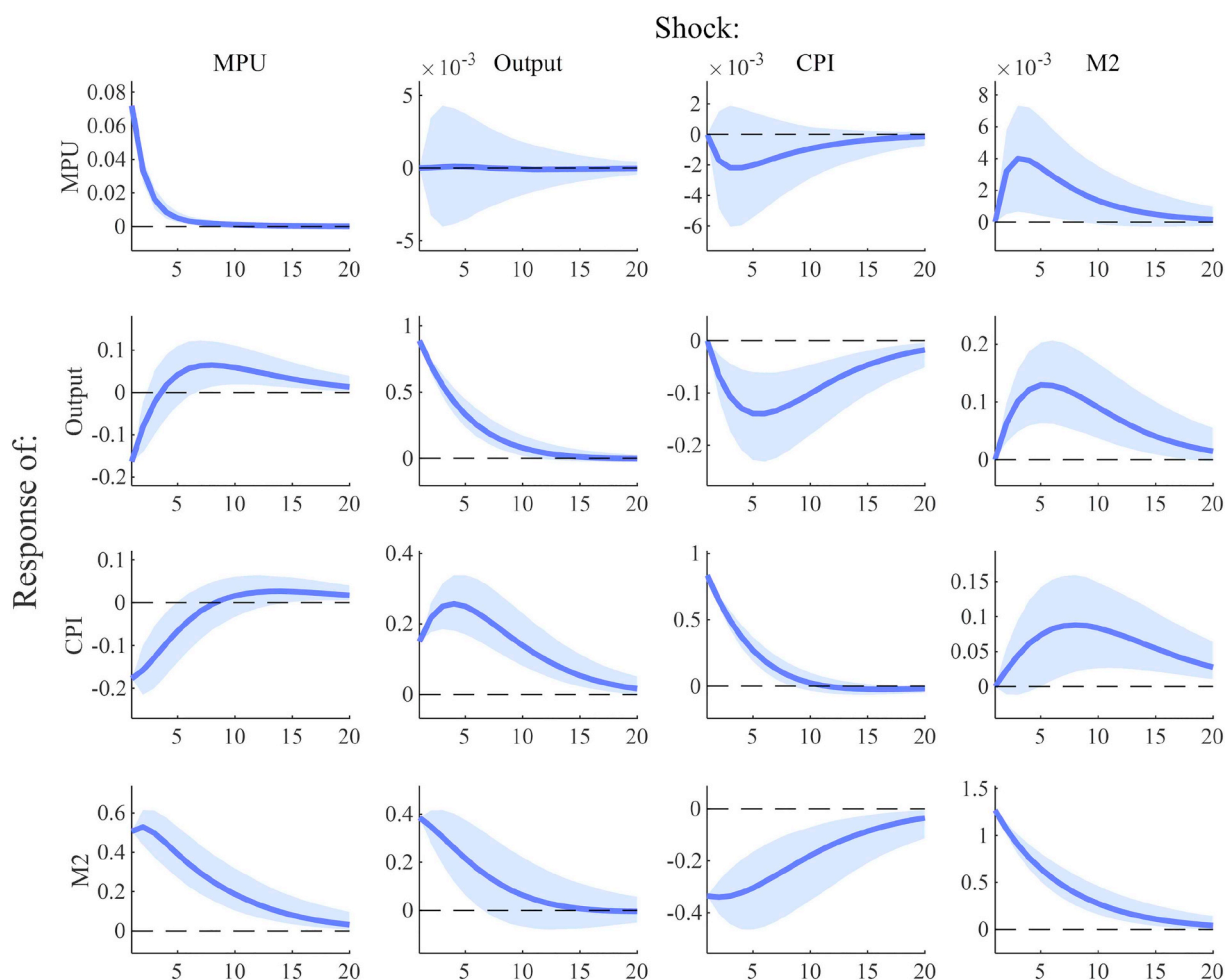


Fig. 2. Impulse responses of macroeconomic variables.

Note: The shaded area is a one standard deviation around the impulse response.

recovery from MPU shock takes longer. To be specific, the output drops by 1.1% at impact of a positive MPU shock, which is larger than a drop of 0.9% under a low credit risk regime. One quarter after the MPU shock, the estimated negative impact gets even stronger and reaches the peak of 2.0%, whereas in low credit risk regime, the impact diminishes and becomes insignificant. Therefore, the expansion of credit can increase the degree of vulnerability in China's economy. At periods of high credit risk, the Chinese government should be especially cautious in steering monetary policy sharply because the induced MPU may exacerbate the downturn in the economy.

Result 3 : Exposure to credit risk magnifies the negative impact of MPU on economic growth.

5. Conclusions

In the context of the Chinese economy, we propose a measure of monetary policy uncertainty (MPU). We estimate China's MPU via a Bayesian MCMC algorithm, and detect a sharp increase in MPU associated with the reaction to the 2008 financial crisis.

Furthermore, we estimate how MPU affects the Chinese economy. We present empirical evidence on the transmission mechanism of MPU from both micro and macro perspectives. Our empirical results suggest that uncertainty in monetary policy hampers the economy in three ways. First, monetary policy uncertainty significantly increases the fraction of non-performing loan in China's commercial banks, resulting in higher credit risk. Second, GDP growth responds negatively to MPU shocks, which is similar to the conclusion drawn from advanced economies in related literatures. Lastly and most importantly, the negative response of GDP growth to MPU shocks is amplified and elongated by high credit risk regimes to which China belonged over the past a few years.

Therefore, a sudden expansion of monetary policy may act like a double-edged sword in stimulating economic growth, because the abrupt change in monetary policy itself may depress economic activities by hiking MPU. To make matters worse, the rise in MPU contributes to higher credit risk, which further amplifies the macroeconomic impact of MPU shocks.

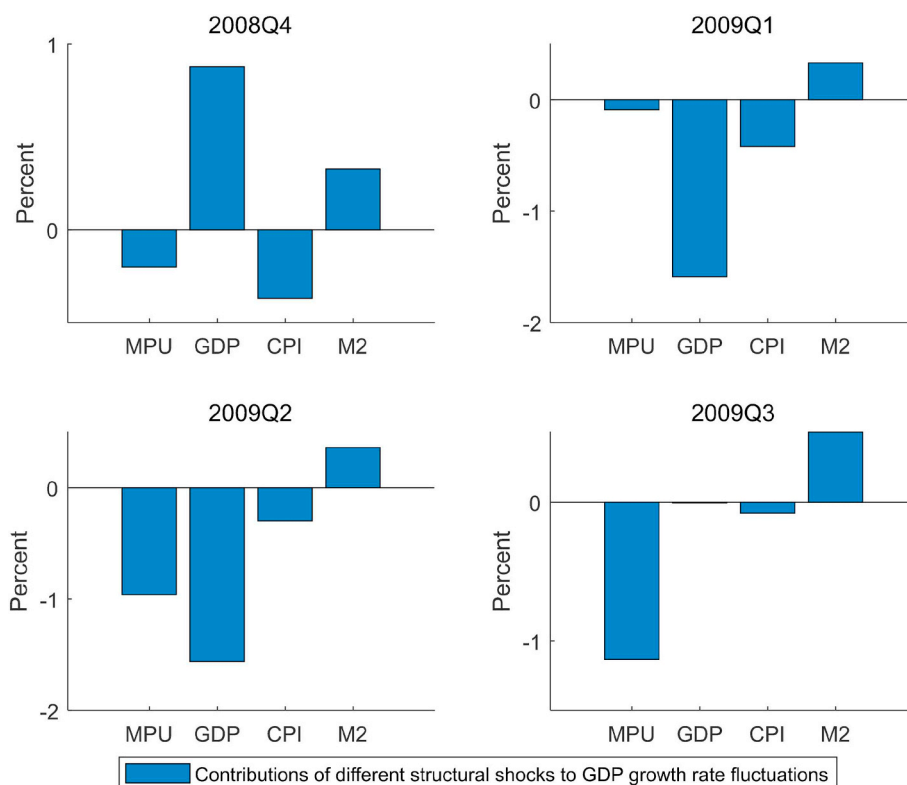


Fig. 3. Historical decomposition for GDP growth during financial crisis.

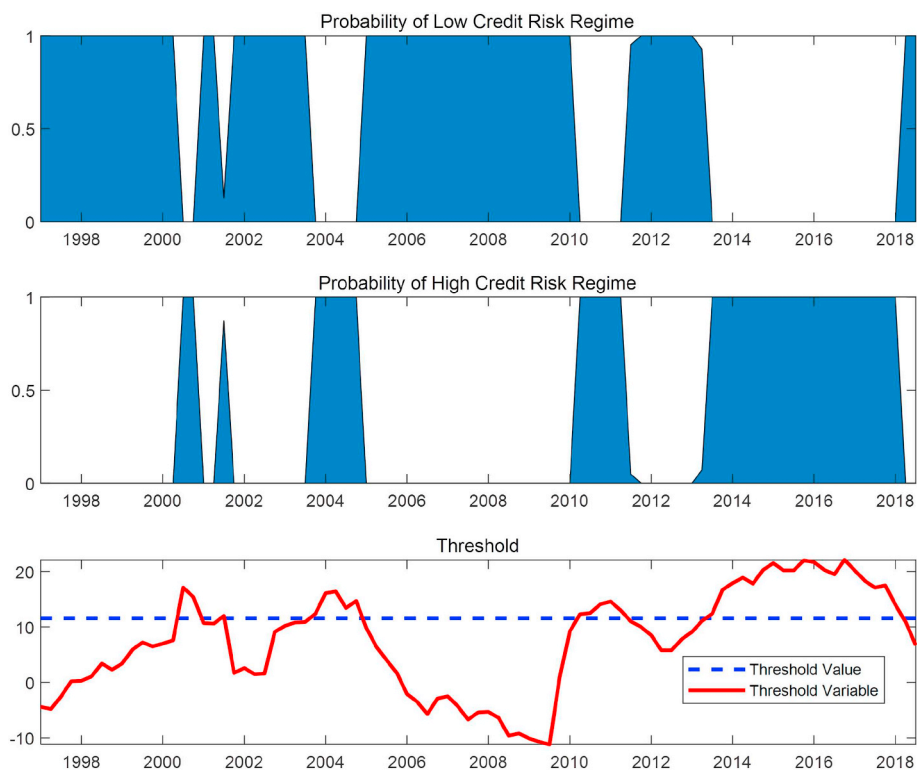


Fig. 4. Regime probabilities.

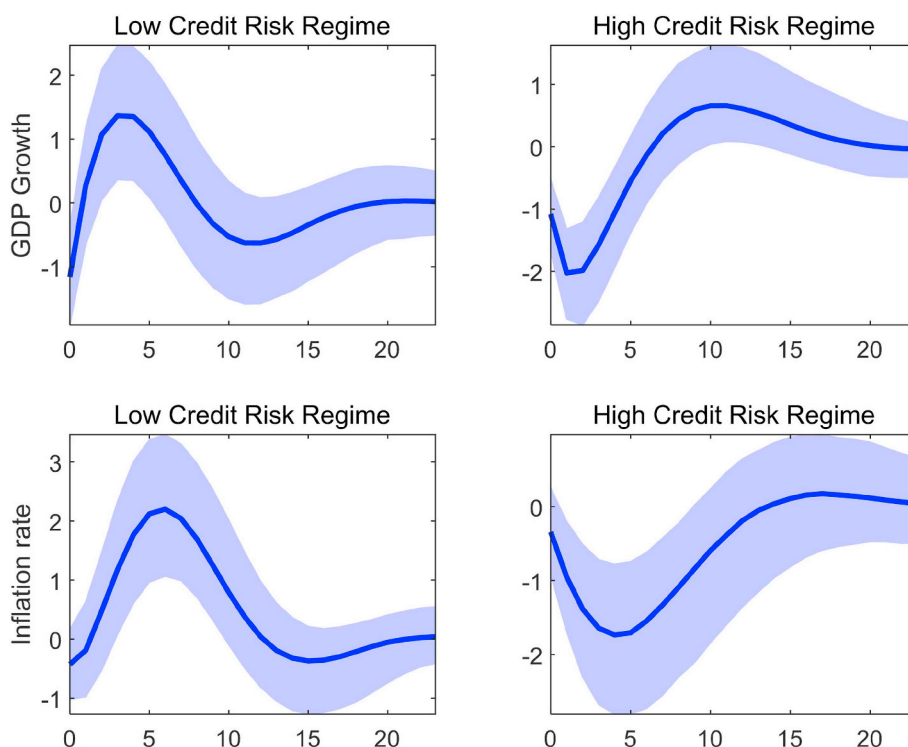


Fig. 5. Impulse responses of MPU shocks under a high (low) credit risk regime.
Note: The shaded area is a one standard deviation around the impulse response.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ememar.2020.100714>.

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